torch.empty_like#

https://pytorch.org/docs/stable/generated/torch.empty_like.html#torch.empty_

Description:

Returns an uninitialized tensor with the same size as input. torch.empty_like(input) is equivalent to torch.empty(input.size(), dtype=input.dtype, layout=input.layout, device=input.device). If torch.use_deterministic_algorithms() and torch.utils.deterministic.fill_uninitialized_memory are both set to True, the output tensor is initialized to prevent any possible nondeterministic behavior from using the data as an input to an operation. Floating point and complex tensors are filled with NaN, and integer tensors are filled with the maximum value. When torch.preserve_format is used: If the input tensor is dense (i.e., non-overlapping strided), its memory format (including strides) is retained. Otherwise (e.g., a non-dense view like a stepped slice), the output is converted to the dense format. input (Tensor) – the size of input will determine size of the output tensor.

Function Signature

```
torch.empty_like(input, *, dtype=None, layout=None,
device=None, requires_grad=False,
memory_format=torch.preserve_format) → Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

dtype (torch.dtype, optional) – the desired data type of returned Tensor. Default: if None, defaults to the dtype of input. layout (torch.layout, optional) – the desired layout of returned tensor. Default: if None, defaults to the layout of input. device (torch.device, optional) – the desired device of returned tensor. Default: if None, defaults to the device of input. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False. memory_format (torch.memory_format, optional) – the desired memory format of returned Tensor. Default: torch.preserve_format.

Code Examples

```
>>> a=torch.empty((2,3), dtype=torch.int32, device = 'cuda')
>>> torch.empty_like(a)
tensor([[0, 0, 0],
        [0, 0, 0]], device='cuda:0', dtype=torch.int32)
```

Notes & Warnings

NOTE

Note If torch.use_deterministic_algorithms() and torch.utils.deterministic.fill_uninitialized_memory are both set to True, the output tensor is initialized to prevent any possible nondeterministic behavior from using the data as an input to an operation. Floating point and complex tensors are filled with NaN, and integer tensors are filled with the maximum value. When torch.preserve_format is used: If the input tensor is dense (i.e., non-overlapping strided), its memory format (including strides) is retained. Otherwise (e.g., a non-dense view like a stepped slice), the output is converted to the dense format.

Detailed Documentation

Documentation

Returns an uninitialized tensor with the same size as input. torch.empty_like(input) is equivalent to torch.empty(input.size(), dtype=input.dtype, layout=input.layout, device=input.device).

If torch.use_deterministic_algorithms() and torch.utils.deterministic.fill_uninitialized_memory are both set to True, the output tensor is initialized to prevent any possible nondeterministic behavior from using the data as an input to an operation. Floating point and complex tensors are filled with NaN, and integer tensors are filled with the maximum value.

When torch.preserve_format is used: If the input tensor is dense (i.e., non-overlapping strided), its memory format (including strides) is retained. Otherwise (e.g., a non-dense view like a stepped slice), the output is converted to the dense format.

input (Tensor) – the size of input will determine size of the output tensor.

dtype (torch.dtype, optional) – the desired data type of returned Tensor. Default: if None, defaults to the dtype of input.

layout (torch.layout, optional) – the desired layout of returned tensor. Default: if None, defaults to the layout of input.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, defaults to the device of input.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

memory_format (torch.memory_format, optional) – the desired memory format of returned Tensor. Default: torch.preserve_format.